

Institute of Actuaries of India

Subject SA6 – Investment

October 2014 Examination

INDICATIVE SOLUTIONS

Solution 1:

i) Guaranteed Annuity Options (GAOs) are options available to holders of certain pension policies. Under these contracts, policyholders contribute either single or regular premiums into a fund managed by the insurer. At retirement, the policyholders have the option to convert the maturity policy proceeds into life annuities at a guaranteed rate provided by the original insurer, or annuitize with another insurer. If the guaranteed annuity rates are more beneficial to the policyholders than the prevailing rates in the market, the insurer has to make up the difference.

Three main risks:

- i. Investment Risk at payout stage- The fall in long-term interest rates – At present, when the interest rates are around 8-9% in India, the GAO (providing 6% guarantee) is out-of-money. But it is difficult to estimate what the interest rates would be in the future (in the next 20-30 years) when these guarantees would be exercised. If they fall significantly, all the options will be deeply in-the-money and insurer would have to bear the cost.
- ii. Longevity Risk during the payout stage- The improvements in life expectancy - this would mean more than expected annuity payments have to be made thereby increasing the total cost of guarantee.
- iii. Anti-selection risk – If the annuity rates are better than 6% then the policyholders may shift to other annuity providers and if the rates are lower than 6% then they may exercise the option. Thus there is a strong anti-selection risk at the time of choosing the annuity provider
- iv. Risk at accumulation stage- The strong equity performance. - The premiums funds are partly invested in equities; however the proportion may be different for unit linked and with-profit contracts. A strong stock market performance would increase the accumulated amounts to which the guarantee would apply.

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ii) Challenges in the Indian pension system.

- Demographic aging: The age structure of the population is changing drastically with increasing life expectancy and declining birth rates. The result of such demographic transition will be a larger proportion of older people in the next few decades
- Changing social mores: Collapse of joint family system coupled with pressures of urbanization and migration are also leading to deterioration in traditional means of support for the elderly.
- Skewed coverage: Existing schemes predominantly cover the organized workers leaving the bulk of the workforce with little access to any formal system of old age income security.
- Inequity in benefits: Generous coverage of the public workers (as far as Pension or other retirement benefits are concerned), compared to the private workers is resulting in further fragmentation of the pension system.
- Pressure on public finances: The spiralling expenditure pattern of the non-contributory, unfunded public pension programs and in some case health cover being extended to retired employees are putting increasing pressure on

government's budgetary allocations. Unless this trend is arrested, these schemes will be financially unsustainable in near future.

- Low returns from provident funds: Conservative investment norms for provident funds, ostensibly to safeguard the workers' interest, may result in inadequate rates of return in the long term from these schemes.
- Under developed private annuity market: Lack of pension annuities and health insurance cover further complicates the old age economic security. Tax on annuities also is a big obstacle in the development of the annuity market.
- Regulatory constraints: Investment guidelines and restrictions on use of derivatives affect product innovation.
- Effect of premature withdrawals: Private schemes are faced with inadequate terminal accumulations, primarily as a result of liberal withdrawals and average returns.
- Need for data on differentiated mortality: Credible data is needed on expected mortality rates of different sub-groups within the diverse Indian population, and on probable improvements in these rates over time. There is lack of data on mortality improvements also.
- Myopia view of the consumers: People do not see a need for annuitizing their savings. Pension products have to be pushed rather than bought voluntarily.
- Investment needs: Long term securities, including government and corporate bonds and mortgage-backed securities, will have to be issued, to reduce reinvestment risk and permit a better matching of assets and liabilities. Debt markets and secondary markets for securities will have to be better developed.

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iii) Three methods of dealing with the type of risks associated with writing financial guarantees.

- Traditional actuarial reserving method – in this method, the insurer sets aside additional capital to ensure that the liabilities under the guarantee will be covered with a higher probability. The liabilities can be estimated using a stochastic simulation approach or through market price of options if similar options are available. For stochastic models the basic idea is to simulate the future using a model of investment returns. These simulations can be used to estimate the distribution of the cost of the guarantee. From this distribution we can compute the amount of initial reserve so that the provision will be adequate, say, 99% of the time.
- Reinsure the liability with another financial institution such as a reinsurance company or an investment bank. In this case the insurance company pays a fee to the financial institution and in return the institution agrees to meet the liability under the guarantee.
- The insurance company can set up a replicating portfolio of traded securities and adjust (or dynamically hedge) this portfolio over time so that at maturity the market value of the portfolio corresponds to the liability under the guaranteed annuity option.
- If the capital market has the appetite for such risks then longevity bonds may be issued. Natural hedge by doing life business and taking mortality risks along with longevity risks may also partly hedge the portfolio but risk due to guarantees on interest rate front will still be there.

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iv) It is like a call option on the bond rather than a put option on the bond since the policyholder can make gains if the interest rate falls and the bond prices go up. He can exercise the option to get a higher notional price of the bond and better annuity rates as explained below.

Assume we have a single premium equity-linked policy. The contract is assumed to mature at T , say, at which date the policyholder is assumed to be age 65. The premium is invested in an account with market value $S(t)$ at time t , where $S(t)$ is a random process.

The market cost of a life annuity of Rs 100 p.a. for a life age 65 is also a random process. Let $a_{65}(t)$ denote this market price.

The policy offers a guaranteed annuity conversion rate of g . This rate determines the guaranteed minimum annuity payment per unit of maturity proceeds of the contract; that is, Rs 100 of the lump sum maturity value must purchase a minimum of Rs 100/ g of annuity.

At maturity the proceeds of the policy are $S(T)$; if the guarantee is exercised this will be applied to maturity amount and if it is treated as a notional bond the value of the bond will be $(S(T)/g) * a_{65}(T)$.

If interest rate has fallen at that time then the value of the bond would increase and the policyholder will have the right to exercise the call option and receive the gain. That gain of the policyholder must be met by the insurer, and will be

$$S(T) / g * a_{65}(T) - S(T)$$

If interest rates remain higher than 6% [$a_{65}(T) < g$] then the guarantee will not be exercised and the cash proceeds will be annuitized without additional cost to insurer. So assuming the policyholder survives to maturity, the value of the guarantee at maturity is

$$S(T) \max [(a_{65}(T) / g - 1), 0]$$

The option will be in-the-money in interest rates fall since the current annuity factor exceeds the guaranteed factor, g .

The market annuity rate $a_{65}(t)$ will depend on the prevailing long-term interest rates, the mortality assumptions used and the expense assumption.

The size of the option liability will be directly proportional to $S(T)$, the amount of proceeds to which guarantee applies. The size of $S(T)$ will depend on the nature of the contract and also the investment returns attributed to the policy.

Assumptions

- Single-premium payments
- No expenses
- Financial risk is independent from mortality risk
- Equity returns are independent of interest rates
- Policyholders behave rationally, i.e. policyholders select the highest annuity payout

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v) A swaption gives its owner the right but not the obligation to enter into an interest rate swap (the terms and conditions of which are set on the trade date) on a specified date in the future. In return for this right, the buyer of the swaption pays a premium to the seller.

Benefits:

- The guaranteed annuity payouts can be matched if we hedge through a swaption wherein we receive the fixed leg of interest and pay the floating rate of interest
- Hedges the buyer against downside risk, but at the same time, retains the advantage of any upside benefits
- Long dated

A portfolio of long-dated receiver interest rate swaptions can be used to hedge interest rate risk under GAOs.

This gives the holder of the swaption the right to enter into a swap where they will receive the predefined (fixed) rate and pay the floating rate.

The holder of a receiver swaption will only enter into this swap if the floating rate is lower than the predefined rate on the exercise date.

By entering into such swaption an institution protects itself against the risk that the interest rates will have fallen when the swaption matures. This is exactly the type of interest rate risk exposure in the guaranteed annuity option.

The swaption strategy creates a static portfolio replicating approach but can have matching issues wherein the value for the swaption may fall short of the maturity value of the pension plan. This means that if the pension premiums have been invested in equities then the number of swaptions has to be adjusted in line with the index movements. During a period of rising equity returns an insurer would have to keep purchasing these swaptions and this would become very expensive as impractical.

It also cannot hedge the risk of increasing longevity and has counterparty risks.

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vi) The main steps to build the ALM are as follows

- Specify the objective of ALM – to specify an asset allocation strategy that meets the solvency requirement within the investment guideline.
- Identify the product features – annuity rate, bonus rate, death benefit, surrender benefit, withdrawals, etc.
- Set the valuation assumptions – longevity, expenses, investment return, inflation, tax.
- Project liability cash flows including reserve and solvency margin.
- Identify the asset models required – nominal and real interest rates, inflation, equity, property.
- Estimate the parameters and assumptions of the asset models, including any correlation of asset classes.
- Simulate the economic series using the stochastic asset models.
- Combine projected liability cash flows and the asset cash flows to produce the net cash flow and the surplus.
- Summarize the results based on VaR, CTE, etc.
- Monitor the model and results at regular intervals.

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vii) The guaranteed annuity option (GAO) at the pay-out stage will not change whether it is under Unit Linked or With Profit Contracts. The difference would be at the accumulation stage and thus the amount at maturity/vesting would differ.

Unit-linked have little insurer discretion and are transparent in the sense that the investment gains and losses are distributed directly to the policyholders' accounts. The policyholders are aware of asset allocation (mainly the debt and equity allocation) and the charges deducted for fund management and other expenses. With the with-profits contracts, there is a discretionary element of smoothening and the investment policy is also opaque which makes them more difficult for the policyholders to analyse. Thus they offer a little more flexibility to insurers in terms of what profits could be passed on to the policyholders and what could be retained by the company. However, bear in mind the PRE.

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viii)

- The asset prices may not be correctly specified. The markets may not be efficient and there may be arbitrage opportunities.
- It will not be possible to rebalance the replicating portfolio on a continuous basis. Instead it has to be rebalanced at discrete intervals.
- There are transaction costs on trading and there may be tax implications also.
- It may be theoretically possible but practically not possible to construct the replicating portfolio due to restrictions on short-selling, restrictions on foreign investments, non-availability of appropriate securities, etc.
- There could be multiple factors which are not identified correctly like ignoring the spread between long and short interest rates, ignoring inflation, etc.
- There may not be a single rate of lending and borrowing or there may be bid-ask spreads.

[3]

ix) Other uses of derivatives by insurance companies:

- Efficient Portfolio management - Use of derivative transactions have the same economic effect as alternative investment in the underlying assets but they are cheaper to implement, eg: in tactical asset allocation decision. Huge stamp duty and dealing costs are reduced but there could be some tracking error.
- Reduce investment risk – by better matching the term, nature and currency of the liabilities. Derivative contracts can be structured as per the needs where trading in the underlying assets is either not possible or too costly.
- Tax management – sell equity-based derivatives rather than underlying equity to avoid crystallising capital gains and incurring a tax liability. More flexible swaps can be used in a similar way, to transfer the economic effects of asset ownership between counterparties without transferring the underlying ownership to avoid tax.
- Innovation - To provide innovative and more attractive structured products to the consumers whose return depends on the derivatives.
- Enhancing returns – alpha (expected outperformance). Alternatively they may enable the generation of what are effectively new asset classes (e.g. market neutral funds) which exhibit low correlation with traditional assets and can therefore improve the overall portfolio risk/return trade-off.

- Solvency management – to protect against the relative fall of liabilities with respect to assets (like interest rate movements, falling equity values).

[3]

x)

- Limited availability of suitable interest rate futures on NSE/ BSE.
- The market is relatively new and favours the sellers. It may not be efficient and prices may be affected by excessive demand.
- Banks and mutual funds are mostly operating in the short to medium-term horizon (sell bonds). There is a shortage of “long only” players (those who buy bonds). However, if insurance companies start participating, there will be players on the long side, which may improve the liquidity. However still it would be difficult to get sellers for long horizons since insurance companies can only use derivatives to hedge and will generally be on the long side of the transaction.
- Irrationality of exercise – policyholders do not always act in the most financially beneficial way.
- Use of derivatives may affect admissibility/ solvency of the company.
- Need to take into account compliance with regulations, guidance notes and PRA (no speculation, ‘window-dressing’ allowed)
- Consider how tax is affected by the use of derivatives
- Accounting issues need to be considered

Regulatory changes

- IRDA has recently allowed the insurers to invest in interest rate derivatives for hedging against interest rate risks in their transactions.
- The insurers are allowed to deal in rupee interest rate derivatives such as forward rate agreements (FRA), interest rate swaps (IRS) and exchange-traded interest rate futures (IRF).
- The insurers are allowed to use interest rate derivatives to hedge for transactions that include reinvestment of maturity proceeds of existing fixed income investments; investment of interest income receivable; and premiums expected to be received on the insurance contracts which are already underwritten in life and pension and annuity business.
- However, an insurer’s dealings in interest rate derivatives cannot exceed an outstanding notional principal amount equivalent to 100% of the book value of its fixed income investments under the policyholders’ fund and the shareholders’ funds taken together.
- While dealing in FRAs and IRS, insurers have to ensure that their counter parties are either commercial banks or primary dealers only.
- All insurers, dealing in interest rate derivatives should also disclose the details of their hedging strategy, accounting policy and nature of outstanding interest rate derivatives in their financial statements. They are also required to disclose the quantum of losses which would be incurred if counter-parties fail to fulfil their obligation under the outstanding interest rate derivative contracts.
- IRS bearing option features are prohibited.

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xi) Products traded in derivatives segment of BSE & NSE

- Index Futures
- Index Options
- Individual Stock Options
- Individual Stock Futures
- Currency Futures
- Interest Rate Futures (IRF) on 10-Year Government of India Security

Trading mechanism on NSE

- The trading system of derivatives at NSE provides a fully automated screen-based trading for all kinds of derivatives products available on NSE on a nationwide basis.
- It supports an anonymous order-driven market, which operates on a time priority/strict price basis.
- It offers great flexibility to users in terms of kinds of orders that can be placed on the terminal. Various time and price-related conditions like Immediate/Cancel, Limit/Market Price, Stop Loss, etc. can be built into an order.
- The trading in derivatives is essentially similar to that of trading of securities in the Capital Market segment.

There are four entities in the trading system of a derivative market:

1. **Trading members:** Trading members can trade either on their own account or on behalf of their clients including participants. They are registered as members with NSE and are assigned an exclusive trading member ID.
2. **Clearing members:** Clearing members carry out confirmation/inquiry of trades and the risk management activities through the trading system. These clearing members are also trading members and clear trade for themselves or/and other.
3. **Professional clearing members:** A clearing member who is not a trading member is known as a professional clearing member (PCM). Typically, banks and custodian become PCMs and clear and settle for their trading members.
4. **Participants:** A participant is a client of trading members like financial institutions. These clients may trade through multiple trading members, but settle their trades through a single clearing member only.

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[52 Marks]

Solution 2:**i) Potential risks in foreign direct investment:**

- There is increased political risk. Political changes may result in changes in how overseas investors are treated.
- There may be business ethics issues that need to be considered.
- There may be difficulties in being able to transfer money out of certain countries
- There might be restrictions on dividend policy, remittance of dividends, repatriation of capital, etc.
- There could be legal risk due to unexpected application of a law or regulation.
- The quality of financial information might be dubious and less timely. Furthermore, the financial information might not be in accordance with international accounting standards.

- Currency complications - there is the risk of adverse exchange rate movements and devaluation.
- The corporation tax rate may be more variable. There may be withholding tax
- The absence of a double taxation relief agreement with an individual country may make many investments uneconomic.
- There may be significant cultural differences, particularly in respect of inducements.
- There may be language and translation delays.
- Time delays and may exist.
- Market regulation may be weaker in some countries than others.
- The overseas markets may have low liquidity or concentrated ownership.
- Ownership restrictions may exist in certain industries (eg airlines, utilities).
- All adds to overall costs and the level of administration expertise required.

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ii) IRR and NPV both take into account the time value of money, which makes them preferable to other methods that do not consider the time value aspect, such as for example the payback methodology. However, in comparison to NPV, IRR has a few drawbacks:

- It gives no indication of materiality. Two projects may appear equally attractive based on their IRR, although the scale is vastly different. The NPV gives indication of such a difference in scale and identifies the most attractive project.
- The IRR cannot take account for variable discount rates, since it only calculates one project return to be compared with an expected market return or a hurdle rate. The NPV, by contrast, can be calculated using a variable discount rate.
- If there are several negative terms (in addition to the initial outlay), there may not be a unique solution for IRR.
- One of the most practical advantages of the NPV, for example in analysing portfolios of projects, is that it is additive. The total value of a portfolio is the sum of its parts. To calculate the portfolio IRR, the individual cash flows would have to be added and then discounted to NPV=0, still giving no indication of the total value of the portfolio, only of its return.
- The IRR, for most people, is more difficult to understand as a concept than the NPV. Also, it is also difficult to calculate without access to a PC if project life is longer than one year.
- The IRR tends to discriminate in favour of short term projects and not those which are more profitable in the long run.

[5]

iii) The Foreign Investment in India is undertaken in accordance with the FDI Policy which is formulated and announced by the Government of India. The Government of India has allowed different channels of Investment in India on basis of the entity of the foreign national.

Who can invest in India?

- A non-resident entity can invest in India, subject to the FDI Policy.
- Erstwhile OCBs which are incorporated outside India and are not under the adverse notice of RBI can make fresh investments under FDI Policy as incorporated non-resident entities.
- An FII may invest in the capital of an Indian Company.

- NRIs resident in Nepal and Bhutan as well as citizens of Nepal and Bhutan are permitted to invest in the capital of Indian companies on repatriation basis.

Entry Routes for Investment in India

- Automatic Route- under the Automatic Route, the foreign investor or the Indian company does not require any approval for investment
- Government Route- under the Government Route, the foreign investor or the Indian company should obtain prior approval of the Government of India or RBI.
- Recent amendments in Indian FDI policy have opened a number of key business sectors to increased foreign investment and in several instances eliminate the need for foreign investors to obtain approval from the Indian government before investing.
 - The RBI has allowed overseas investors, including foreign portfolio investors (FPIs) and NRIs, to invest up to 26 per cent in insurance and related activities via the automatic route.
 - The RBI has allowed a number of foreign investors to invest, on repatriation basis, in non-convertible/ redeemable preference shares or debentures which are issued by Indian companies and are listed on established stock exchanges in the country.

Types of Instruments to invest in India

- Equity shares
 - Issuance of fresh shares by the company
 - Acquisition by way of transfer of existing shares by person resident in or outside India
 - Issue of Rights / Bonus shares
 - Issue of shares under Employees Stock Option Scheme
 - Issue of shares by Indian Companies under ADR / GDR
 - Acquisition of shares under Scheme of Merger / Amalgamation
- Preference shares
- Debentures

[6]

iv) The relative advantages and disadvantages of the use of a money market hedge versus exchange traded derivatives:

- A money market hedge can provide preferential access to the short term money markets in the base and counter currency zones and be a cost effective substitute for a forward agreement.
- However, it is difficult to reverse quickly and is cumbersome to establish. Exchange traded derivatives such as futures and foreign exchange options offer a rapid way of creating a hedge and are easily closed out.
- The fixed contract sizes for exchange traded products mean that it is often impossible to achieve a perfect hedge and some gain or loss on the unhedged element of the underlying or the derivative will be carried.
- Also, given that exchange traded derivatives are priced in a separate market to the underlying, there may be discrepancies in the movements of each and the observed delta may not equal one. This basis risk is minimized by choosing short maturity derivatives.

- Currency hedging using the derivatives market may also involve significant cash-flows in meeting and maintaining the margin requirements of the exchange.
- Unlike futures, currency options will entail the payment of a premium which may be an expensive way of eliminating the risk of an adverse currency movement.
- In this case, relatively infrequent flows occur; therefore the simplest solution is to engage in a money market hedge than direct recourse to the forex market.
- However the administrative and management costs in setting up the necessary loans and deposits are a significant consideration.

Effect on company's cost of capital:

- Currency risk forms part of a firm's exposure to market risk and will impact on a firm's cost of capital through its beta value and, as a result, its equity cost of capital. The extent to which this is significant depends on the exposed volume of currency transactions and the correlation of the currency with the market. If a given currency has the same correlation with the market as the company, removing currency risk will have no impact on the firm's overall exposure to market risk and no impact on the firm's cost of capital. The greater the difference in the relative correlations the higher the potential improvement in shareholder value from hedging.
- The impact on the cost of debt is more complex. The most significant impact is through the firm's exposure to default risk. If currency transactions are significant and the foreign currency is highly correlated with the domestic currency then the impact is unlikely to be significant and the gains from hedging modest. Where the degree of correlation is low or indeed negative then eliminating currency risk may significantly alter the firm's default risk.

[6]

v) a. The operating margin under base case

If we consider the company to be based out of USA and that it has attempted to make an investment in India then the operating profits at USA would not be impacted due to the change in the exchange rate. However the Indian denominated operating profits would get impacted by 10% up and down in case the exchange varies and the change in profit due to such currency movement would be to the tune of Rs 50 crores (500/10) up or down.

However we may also look at the case from the Indian entity point of view wherein the impact on the USA operating profits only would be considered due to the exchange rate movement in which case the impact would be to the tune of Rs 80 crores up or down as shown below.

Base Case

Item	USD	INR	Total
Revenue	1,000	200	1,200
Cost	200	700	900
Operating Profits			300
Operating margin (as % of sales)		25%	
Operating margin if INR depreciates by 10%			
Item	USD	INR	Total
Revenue	900	200	1,100
Cost	180	700	880
Operating Profits			220
Operating margin		20%	
Operating margin if INR appreciates by 10%			
Item	USD	INR	Total
Revenue	1100	200	1,300
Cost	220	700	920
Operating Profits			380
Operating margin		29%	
			[5]

(b) Risks:

Basis risk – the value of forward contract may not match the realised exchange rate under the contract;

Operational risk – the company is a technology firm not a financial institution – it may not have the necessary expertise and processes to manage derivative contracts;
c

Counterparty risk – Being an OTC contract it has counterparty risks

Limitations: Uncertainty of the magnitude and timing of the underlying cashflows; 6 mth maturity seems arbitrary; revenue and costs are incurred throughout the year – forward contract has a fixed maturity date, transaction costs will have to be taken into consideration [5]

vi) a.] Interest rate risk – fall in interest rates increases the present value of future payments.

Wage inflation risk – higher wage inflation means higher benefit payment, also increase in expected future inflation could lead to higher provision for future benefits

Termination rates – lower than expected termination rates would lead to increase in provision; higher than expected termination rates could force investments to be liquidated earlier than expected and may be on unfavourable terms

Risks due to significant change in the Age and Income Profile of the group.

[4]

(b) Cash management bills: short-term instruments; issued by GoI to meet temporary mismatches in the cash flow of the Government. Maturities – less than 91 days; issued at discount redeemed at par,

Treasury bills: Money market instruments, issued for three tenors 91 days, 182 days, 364 days. Also issued at discount redeemed at par.

Dated Government securities: Long term government bonds; fixed or floating interest rates, usually half yearly interest payments; Tenor upto 30 years; There are no Zero Coupon bonds or Capital Indexed bonds in circulation at the moment;

[3]

(c) Construction of a hedged portfolio requires producing best-estimate liability cash-flow profile; this should take into consideration

- the terms of gratuity scheme,
- profile plan members
- best estimate termination rates
- best estimate wage inflation

The sensitivity of present value of liability cash-flows to change in interest rate, usually in terms of Dollar Duration (DV01), should be calculated to facilitate hedging.

These DV01s can be grouped into chosen maturity brackets. For example, 0-1, 2-5, 5-10, 10-15, 15-25, 25+ (these brackets can be finer or broader depending on how accurate the hedge needs to be and also availability of assets to hedge).

Similarly, DV01s, in each bracket, of available assets (CMBs, T-bills and Dated bonds) should be calculated per unit nominal.

The sensitivity in the longest maturity bracket should be hedged with the bonds of appropriate maturity and face value such that the net DV01 for the bracket is zero.

In theory, the process can be repeated with earlier maturities such that the net DV01 is zero for each chosen bracket.

Due consideration should be given to the DV01 of coupon payments on coupon paying bonds while repeating this exercise.

Additional and residual risks:

Basis risk, credit risk, reinvestment risk,

Gaps in bond maturities, excess hedging in initial years, unhedged inflation risk.

[8]

[48 Marks]
